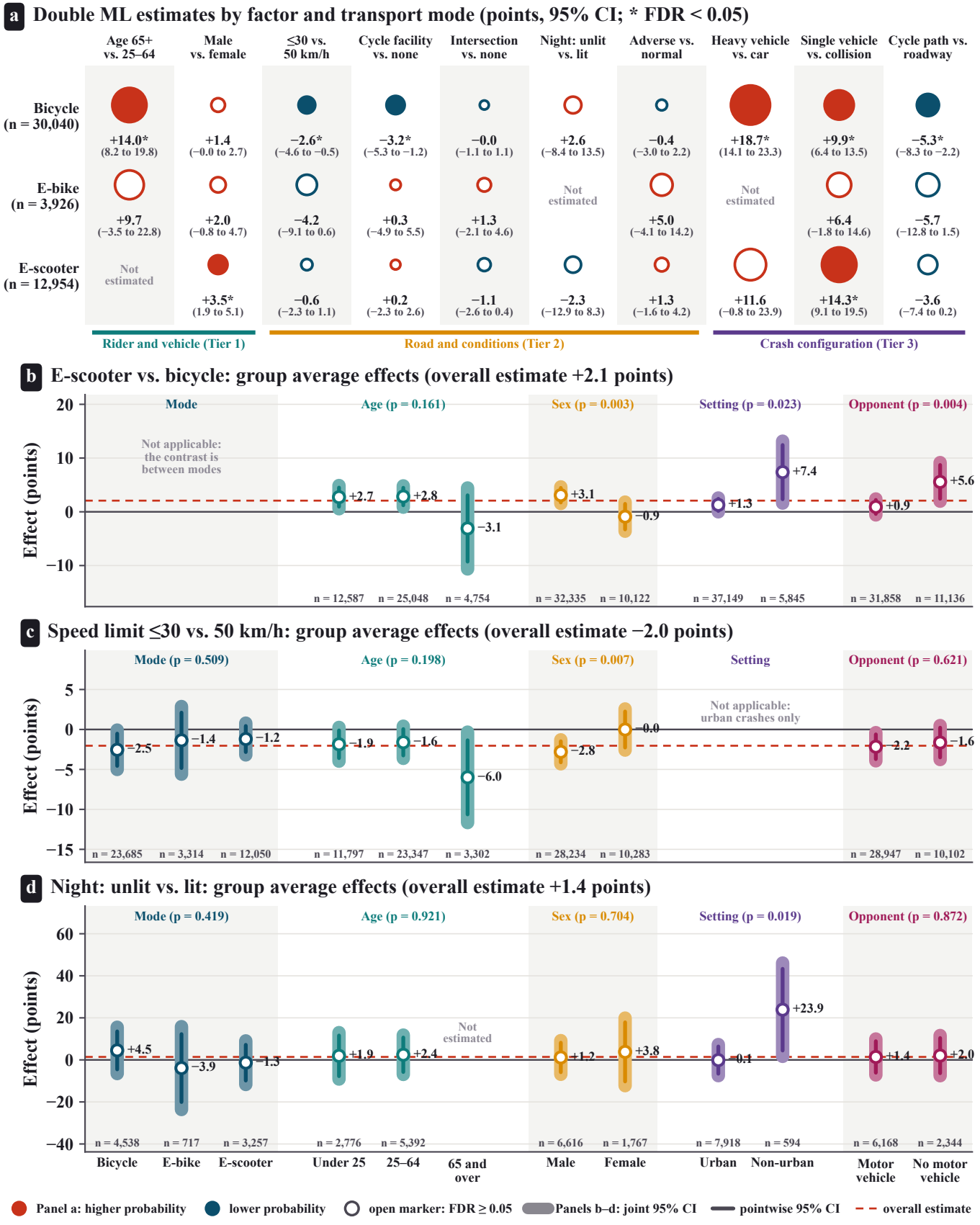
a Double ML estimates by factor and transport mode (points, 95% CI; * FDR < 0.05)
Age 65+ vs. 25–64
Male vs. female
≤30 vs. 50 km/h
Cycle facility vs. none
Intersection vs. none
Night: unlit vs. lit
Adverse vs. normal
Heavy vehicle vs. car
Single vehicle vs. collision
Cycle path vs. roadway
Bicycle (n = 30,040)
+14.0* (8.2 to 19.8)
+1.4 (−0.0 to 2.7)
−2.6* (−4.6 to −0.5)
−3.2* (−5.3 to −1.2)
−0.0 (−1.1 to 1.1)
+2.6 (−8.4 to 13.5)
−0.4 (−3.0 to 2.2)
+18.7* (14.1 to 23.3)
+9.9* (6.4 to 13.5)
−5.3* (−8.3 to −2.2)
E-bike (n = 3,926)
+9.7 (−3.5 to 22.8)
+2.0 (−0.8 to 4.7)
−4.2 (−9.1 to 0.6)
+0.3 (−4.9 to 5.5)
+1.3 (−2.1 to 4.6)
Not estimated
+5.0 (−4.1 to 14.2)
Not estimated
+6.4 (−1.8 to 14.6)
−5.7 (−12.8 to 1.5)
E-scooter (n = 12,954)
Not estimated
+3.5* (1.9 to 5.1)
−0.6 (−2.3 to 1.1)
+0.2 (−2.3 to 2.6)
−1.1 (−2.6 to 0.4)
−2.3 (−12.9 to 8.3)
+1.3 (−1.6 to 4.2)
+11.6 (−0.8 to 23.9)
+14.3* (9.1 to 19.5)
−3.6 (−7.4 to 0.2)
Rider and vehicle (Tier 1)
Road and conditions (Tier 2)
Crash configuration (Tier 3)
b E-scooter vs. bicycle: group average effects (overall estimate +2.1 points)
Effect (points)
Mode
Not applicable: the contrast is between modes
Age (p = 0.161)
+2.7
+2.8
−3.1
n = 12,587
n = 25,048
n = 4,754
Sex (p = 0.003)
+3.1
−0.9
n = 32,335
n = 10,122
Setting (p = 0.023)
+1.3
+7.4
n = 37,149
n = 5,845
Opponent (p = 0.004)
+0.9
+5.6
n = 31,858
n = 11,136
c Speed limit ≤30 vs. 50 km/h: group average effects (overall estimate −2.0 points)
Effect (points)
Mode (p = 0.509)
−2.5
−1.4
−1.2
n = 23,685
n = 3,314
n = 12,050
Age (p = 0.198)
−1.9
−1.6
−6.0
n = 11,797
n = 23,347
n = 3,302
Sex (p = 0.007)
−2.8
0.0
n = 28,234
n = 10,283
Setting
Not applicable: urban crashes only
Opponent (p = 0.621)
−2.2
−1.6
n = 28,947
n = 10,102
d Night: unlit vs. lit: group average effects (overall estimate +1.4 points)
Effect (points)
Mode (p = 0.419)
+4.5
−3.9
−1.3
n = 4,538
n = 717
n = 3,257
Age (p = 0.921)
Not estimated
+1.9
+2.4
n = 2,776
n = 5,392
Sex (p = 0.704)
+1.2
+3.8
n = 6,616
n = 1,767
Setting (p = 0.019)
−0.1
+23.9
n = 7,918
n = 594
Opponent (p = 0.872)
+1.4
+2.0
n = 6,168
n = 2,344
Bicycle
E-bike
E-scooter
Under 25
25–64
65 and over
Male
Female
Urban
Non-urban
Motor vehicle
No motor vehicle
Panel a: higher probability
lower probability
open marker: FDR ≥ 0.05
Panels b–d: joint 95% CI
pointwise 95% CI
overall estimate